

Pipeline Forecasting Engine

Weighted forecast modeling built on historical win rates and stage velocity, with scenario what-ifs and rep-level roll-ups.

Python	FastAPI	Postgres	scikit-learn
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VERSION	STATUS	CADENCE	RUNTIME	OWNER	UPDATED
2.2.0	Live	Nightly	Python 3.12	RevOps Eng	Sep 2026

1. Overview

The Pipeline Forecasting Engine produces a weighted revenue forecast from open opportunity data rather than from rep-entered commit categories. Each open deal is scored against historical win rates for its stage, segment, and deal size band, then adjusted by how long it has sat relative to the stage velocity of deals that eventually closed. The result is a probability-weighted amount per deal that rolls up to rep, team, and region.

Two things make the output defensible in a forecast call. First, every number is traceable: the API returns the contributing deals and the factors applied to each, so a disputed number resolves by inspection instead of argument. Second, scenario what-ifs run against the same model — pull a deal forward a stage, slip a close date, change an amount — without mutating CRM data, so managers can test a story before committing to it.

2. Forecast pipeline

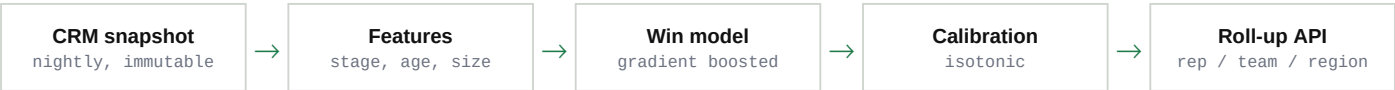


Figure 1 — The nightly path. Scenario requests re-enter at the calibration stage and reuse the trained model without a retrain.

Snapshots are the foundation. Each night the engine writes an immutable copy of every open opportunity; forecasts are computed against snapshots, never live CRM state. That makes historical forecasts reproducible and drift attributable — when this week's number moves, the engine can say which deals moved it.

3. System layers

Snapshot store	Append-only Postgres tables, one partition per day, with a 25-month retention window.
Feature builder	Derives stage age, velocity percentile, activity recency, amount band, and segment.
Win model	Gradient-boosted classifier per segment, retrained weekly on 18 months of history, with isotonic calibration so scores match observed close rates.
Scenario engine	Applies in-memory deal overrides and recomputes a roll-up without writing anything.
API	FastAPI service with roll-up, drill-down, scenario, and accuracy-history endpoints.

4. Weighting inputs

FACTOR	EFFECT ON PROBABILITY	SOURCE
Stage	Primary term; base rate by stage and segment.	Historical closed-won rate, 18-month window.
Stage age	Decays probability past the p75 velocity mark.	Time-in-stage from snapshot history.
Amount band	Large deals discounted; they slip more often.	Win rate by decile of deal size.
Activity recency	Silence beyond 14 days applies a penalty.	Task and meeting timestamps.

5. Scenario API

```
POST /v2/forecast/scenario
{
  "period": "2026-Q4",
  "scope": { "team": "enterprise-west" },
  "overrides": [
    { "opportunity": "006Ab0001", "stage": "negotiation" },
    { "opportunity": "006Ab0742", "close_date": "2027-01-15" },
    { "opportunity": "006Ab0913", "amount": 185000 }
  ],
  "explain": true           // return per-deal factor contributions
}

// response
{ "weighted": 4128500, "baseline": 3874000, "delta": 254500,
  "by_rep": [{ "rep": "a.patel", "weighted": 962000, "deals": 14 }],
  "explain": [{ "opportunity": "006Ab0001", "p": 0.62, "was": 0.34,
    "factors": { "stage": +0.31, "age": -0.04, "size": +0.01 } } ] }
```

Listing 1 — Overrides live only in the request; nothing is written back to the CRM.

6. Model training and validation

- **Training.** Weekly retrain per segment on an 18-month rolling window, with deals from the most recent 45 days held out to avoid leaking outcomes that are not yet settled.
- **Validation.** Each candidate model must beat the incumbent on log loss and on calibration error before promotion; a regression on either metric blocks the release and keeps the prior model serving.
- **Guardrails.** Segments with fewer than 200 closed deals in the window fall back to a pooled model rather than fitting on thin data, and the API marks those roll-ups as low confidence.

7. Forecast accuracy

Measured against actual closed-won for the four most recent quarters, comparing the engine's weighted number to the rep-entered commit at the same point in the quarter.

HORIZON	ENGINE MAPE	REP COMMIT MAPE	CALIBRATION ERROR
90 days out	14.2%	27.6%	0.041
45 days out	8.7%	16.1%	0.033
14 days out	4.1%	7.9%	0.021

8. Operations

- The nightly job completes in roughly 11 minutes for 40k open opportunities; a failure leaves the prior snapshot serving rather than producing a partial forecast.
- Scenario requests are synchronous and cached per scope for five minutes, so a manager iterating in a forecast call sees sub-second responses after the first query.
- Accuracy is tracked continuously and alerted on: a calibration error above 0.08 for two consecutive weeks pages the owning team, since that usually signals a process change in the CRM rather than a model fault.

9. Roadmap

RELEASE	PLANNED WORK	TARGET
2.3	Per-deal risk narratives generated from the dominant negative factors.	Q4 2026
2.4	Multi-currency roll-ups with period-locked FX rates.	Q1 2027
3.0	Bookings-to-revenue bridge covering ramped and multi-year contracts.	Q2 2027